

## 39.2 Diagnostics

A fitted generalized linear model can be wrong in three ways: the linear predictor may omit a term or use a covariate on the wrong scale, the link may be wrong, or the variance function may be wrong. A single number — the deviance, say — cannot tell these apart, and for ungrouped binary data it cannot even tell that something is wrong (Proposition 35.4.1(b)). What follows is a set of plots and tests, each aimed at one of the three, and a worked case in which they give different answers.

### 39.2.1 Residuals in practice

The three residual types of Definition 34.5.2 differ in what they standardize away. Their common first-order behaviour is the generalized linear version of Eq. 20.1.2.

#### Proposition 39.2.1 — What the standard diagnostics measure

Let a generalized linear model be fitted by maximum likelihood, let  $\hat{\mathbf{W}} = \text{diag}\{w_i h'(\hat{\eta}_i)^2 / V(\hat{\mu}_i)\}$  be the working weights,  $\mathbf{H} = \hat{\mathbf{W}}^{1/2} \mathbf{X} (\mathbf{X}^\top \hat{\mathbf{W}} \mathbf{X})^{-1} \mathbf{X}^\top \hat{\mathbf{W}}^{1/2}$  the weighted hat matrix and  $h_{ii}$  its diagonal.

- (a) (*Moments of the Pearson residuals.*) To first order in the expansion behind Theorem 34.5.1,  $E(r_i^P) \approx 0$  and  $\text{Var}(r_i^P) \approx 1 - h_{ii}$ , and the standardized residual  $r_i^P / \sqrt{1 - h_{ii}}$  has approximate variance one. Moreover  $\sum_i h_{ii} = p$ , so  $h_{ii}$  averages  $p/n$ .
- (b) (*Level of a simulated envelope.*) Let  $|r|_{(1)} \leq \dots \leq |r|_{(n)}$  be the ordered absolute residuals, and let  $B$  independent data sets be generated from the fitted model and refitted, giving vectors of ordered absolute residuals whose  $k$ th entries are  $R_k^{(1)}, \dots, R_k^{(B)}$ . If the model is correct and the fitted parameters are the true ones, and the  $B + 1$  values  $|r|_{(k)}, R_k^{(1)}, \dots, R_k^{(B)}$  are exchangeable with no ties, then

$$\Pr\{|r|_{(k)} < \min_b R_k^{(b)} \text{ or } |r|_{(k)} > \max_b R_k^{(b)}\} = \frac{2}{B+1} \quad (39.2.1)$$

for each fixed  $k$ . The probability that *some*  $k$  leaves the band is larger than  $2/(B+1)$ , and no better than the union bound  $\min\{1, 2n/(B+1)\}$  without further argument.

- (c) (*Influence.*) The one-step deletion results of Theorem 20.3.1 hold with  $\mathbf{X}$  replaced by  $\hat{\mathbf{W}}^{1/2} \mathbf{X}$  and  $\mathbf{y}$  by the working response  $\mathbf{z}$  of Eq. 34.4.1, giving the Cook-type measure  $C_i = (r_i^P)^2 h_{ii} / \{p(1 - h_{ii})^2\}$  of Definition 20.4.1. The exact deletion identities do not hold, because removing a case changes the weights as well.
- (d) (*Checking the link.*) Let  $\hat{\eta}_i$  be the fitted linear predictor and refit the model with  $\hat{\eta}_i^2$  added as a covariate. Under the fitted link the extra coefficient is zero, so the deviance drop is asymptotically  $\chi^2(1)$ ; a significant drop says the link is wrong, not which link is right.
- (e) (*Checking a covariate: two plots.*) The partial residual for covariate  $j$  on the link scale is  $u_{ij} = \hat{\beta}_j x_{ij} + (y_i - \hat{\mu}_i) / h'(\hat{\eta}_i)$ . Its weighted least squares regression on  $[\mathbf{1}, \mathbf{x}_j]$  with weights  $\hat{\mathbf{W}}$  has slope  $\hat{\beta}_j$  exactly, so a smooth of  $u_{ij}$  against  $x_{ij}$  that departs from a straight line indicates a wrong scale for  $x_j$ , as in Proposition 20.1.3.

The added-variable plot is the other half of the pair: let  $\tilde{\mathbf{z}}$  and  $\tilde{\mathbf{x}}_j$  be the residuals of the working response  $\mathbf{z}$  and of  $\mathbf{x}_j$  after  $\hat{\mathbf{W}}$ -weighted regression on the remaining columns  $\mathbf{X}_{-j}$ . Then the  $\hat{\mathbf{W}}$ -weighted regression of  $\tilde{\mathbf{z}}$  on  $\tilde{\mathbf{x}}_j$  has slope exactly  $\hat{\beta}_j$ , so curvature or a single dominating point in that plot is a statement about  $\hat{\beta}_j$  itself.

- (f) (*Checking the variance function.*) Suppose the assumed variance function is correct and each observation carries enough information for  $r_i^P$  to be approximately normal: grouped data with large group sizes, or counts with means that are not small. (This fails for ungrouped binary data, where  $r_i^P$  takes two values and the constant below is wrong.) Then  $E|r_i^P| \approx \sqrt{2/\pi}$  for every  $i$ , so a plot of  $|r_i^P|$  against  $\hat{\eta}_i$  should be level. A systematic trend indicates a variance function of the wrong shape; a level plot at the wrong height indicates a dispersion  $\phi \neq 1$ .

*Proof.*

- (a) The expansion used in Theorem 34.5.1 gives, to first order,  $\hat{\beta} - \beta \approx (\mathbf{X}^\top \mathbf{W} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{W} \mathbf{D}^{-1} (\mathbf{y} - \boldsymbol{\mu})$  and hence  $\hat{\boldsymbol{\mu}} - \boldsymbol{\mu} \approx \mathbf{D} \mathbf{X} (\hat{\beta} - \beta)$ , where  $\mathbf{D} = \text{diag}(d\mu_i/d\eta_i)$ . Put  $\mathbf{u} = \mathbf{W}^{1/2} \mathbf{D}^{-1} (\mathbf{y} - \boldsymbol{\mu})$ , whose entries are the Pearson residuals evaluated at the true parameter, with  $\text{Cov}(\mathbf{u}) = \mathbf{I}$  by the definition of  $\mathbf{W}$ . Then

$$\begin{aligned} \mathbf{W}^{1/2} \mathbf{D}^{-1} (\mathbf{y} - \hat{\boldsymbol{\mu}}) &\approx \mathbf{u} - \mathbf{W}^{1/2} \mathbf{X} (\mathbf{X}^\top \mathbf{W} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{W}^{1/2} \mathbf{u} \\ &= (\mathbf{I} - \mathbf{H}) \mathbf{u}, \end{aligned}$$

so  $E(r_i^P) \approx 0$  and  $\text{Var}(r_i^P) \approx (\mathbf{I} - \mathbf{H})_{ii} = 1 - h_{ii}$ . The trace identity is Proposition 6.3.4 applied to  $\mathbf{H}$ , which is a projection matrix.

- (b) The event that  $|r|_{(k)}$  is the strict minimum or the strict maximum of the  $B + 1$  exchangeable values has probability  $2/(B + 1)$ , because each of the  $B + 1$  values is equally likely to be the smallest and equally likely to be the largest, and with no ties these two events are disjoint for  $B \geq 1$ . The final sentence is the usual multiplicity statement: the events for different  $k$  are highly dependent, so the probability that some  $k$  leaves the band lies between  $2/(B + 1)$  and  $\min\{1, 2n/(B + 1)\}$  by the union bound, and is not  $2/(B + 1)$ .
- (c) Theorem 20.3.1 is an algebraic identity about a least squares fit. At convergence the generalized linear fit *is* a weighted least squares fit of  $\mathbf{z}$  on  $\mathbf{X}$  with weights  $\hat{\mathbf{W}}$  (Theorem 34.4.1), so the identity applies to that problem, which is what the substitution says. It is only one step of the algorithm, since refitting without case  $i$  changes  $\hat{\boldsymbol{\mu}}$  and therefore  $\hat{\mathbf{W}}$ .
- (d) The enlarged model contains the fitted one at coefficient zero and adds one parameter, so Theorem 34.5.5 gives the  $\chi^2(1)$  limit under the smaller model. Note that  $\hat{\eta}_i$  is itself estimated. Because  $\hat{\eta}^2$  is a function of the fitted values alone, the normal-theory analogue is Theorem 14.7.2, which makes the corresponding  $F$  test *exact*; here the response is not normal, the limit is the asymptotic one, and the effect of estimating  $\hat{\eta}_i$  is of smaller order under the conditions of Theorem 34.5.1.
- (e) The working residual is  $(y_i - \hat{\mu}_i)/h'(\hat{\eta}_i) = z_i - \hat{\eta}_i$  by Eq. 34.4.1. The weighted least squares fit of  $\mathbf{z}$  on  $\mathbf{X}$  has residual vector  $\mathbf{z} - \hat{\boldsymbol{\eta}}$  orthogonal to  $\mathbf{X}$  in the  $\hat{\mathbf{W}}$  inner product, so  $\mathbf{u}_j = \hat{\beta}_j \mathbf{x}_j + (\mathbf{z} - \hat{\boldsymbol{\eta}})$  regressed on  $[\mathbf{1}, \mathbf{x}_j]$  in that inner product returns  $\hat{\beta}_j$  exactly, by Proposition 20.1.3(a) applied in the  $\hat{\mathbf{W}}$  metric (Theorem 6.9.1). For the added-variable statement, at convergence the fit is the weighted least squares regression of  $\mathbf{z}$  on  $\mathbf{X}$  with weights  $\hat{\mathbf{W}}$  (Theorem 34.4.1), which is an ordinary least squares problem in the  $\hat{\mathbf{W}}$  inner product; Theorem 6.6.2 applied in that inner product gives the claim.

- (f) By (a),  $r_i^P$  has approximate mean zero and variance  $1 - h_{ii} \approx 1$  when the leverages are small. Its approximate normality is the hypothesis of the statement, and it is a statement about the single observation  $Y_i$ , not about  $\hat{\beta}$ : Theorem 34.5.1 concerns the estimate and does not deliver it. For  $Z \sim N(0, 1)$ ,  $E|Z| = \sqrt{2/\pi}$ . If instead  $\text{Var}(Y_i) = \phi V(\mu_i)/w_i$  with  $\phi \neq 1$ , every  $r_i^P$  is inflated by  $\sqrt{\phi}$  and the plot is level at  $\sqrt{2\phi/\pi}$ . ■

*Remark* (Why absolute residuals). A quantile plot of signed residuals asks two questions at once, and for a discrete response the answer to the first is dominated by the discreteness. Plotting  $|r|_{(k)}$  against  $\Phi^{-1}\{(k + n - 1/8)/(2n + 1/2)\}$ , the expected order statistics of  $|Z|$ , asks only about magnitude, which is what an outlier or an inflated dispersion changes. The envelope turns the picture into a parametric bootstrap test (Proposition 23.3.3), of only approximate level because the data sets are generated at  $\hat{\beta}$ .

## 39.2.2 A model that fails, and why

### Example. (Days of television news)

The 1996 American National Election Studies survey asked its 944 respondents on how many of the past seven days they had watched the television news. Treating the answer as a binomial count out of seven and fitting a logit model in age, education and income gives a deviance of 4447.3 on 940 degrees of freedom and a Pearson statistic of 3598.2, so  $\hat{\phi} = 3.828$ : the model is not close to fitting.

Panel (a) of Figure 39.2.1 is the half-normal plot with an envelope from  $B = 19$  simulated data sets, refitted each time. Every one of the 944 ordered residuals lies above the band — not a little too large in the tail but too large everywhere, the signature of a dispersion problem rather than of outliers.

The targeted checks of Proposition 39.2.1 say where the trouble is not. Adding  $\hat{\eta}^2$  drops the deviance by 0.31 on one degree of freedom,  $p = 0.580$ : the logit link is not the problem. Adding a quadratic in age drops it by 0.03, and panel (a) of Figure 39.2.2 shows the partial residual for age with a smooth that is mildly concave but close to the fitted line, so the scale of age is not the problem either. The leverages are all tiny — the largest 0.0144 against an average of 0.0042 — and the largest Cook-type distance is 0.0970.

What is left is the variance function, and panel (b) makes the case. The local means of  $|r^P|$  in ten equal bins of the linear predictor run from 1.25 to 1.91, against a reference height of  $\sqrt{2/\pi} \approx 0.80$ , and they drift upward: the regression of  $\log(r^P)^2$  on the linear predictor has slope 0.381 with  $t = 4.21$ , so the excess variance is not even constant. The binomial model treats a respondent's seven days as seven independent trials with a common probability, and they are nothing of the kind: 161 respondents watched on none of the seven days and 288 on all seven. Habits are not coin flips.

The repair is a variance function with a second parameter. A beta-binomial model, in which each respondent has a latent probability from a beta distribution with mean  $\pi_i$

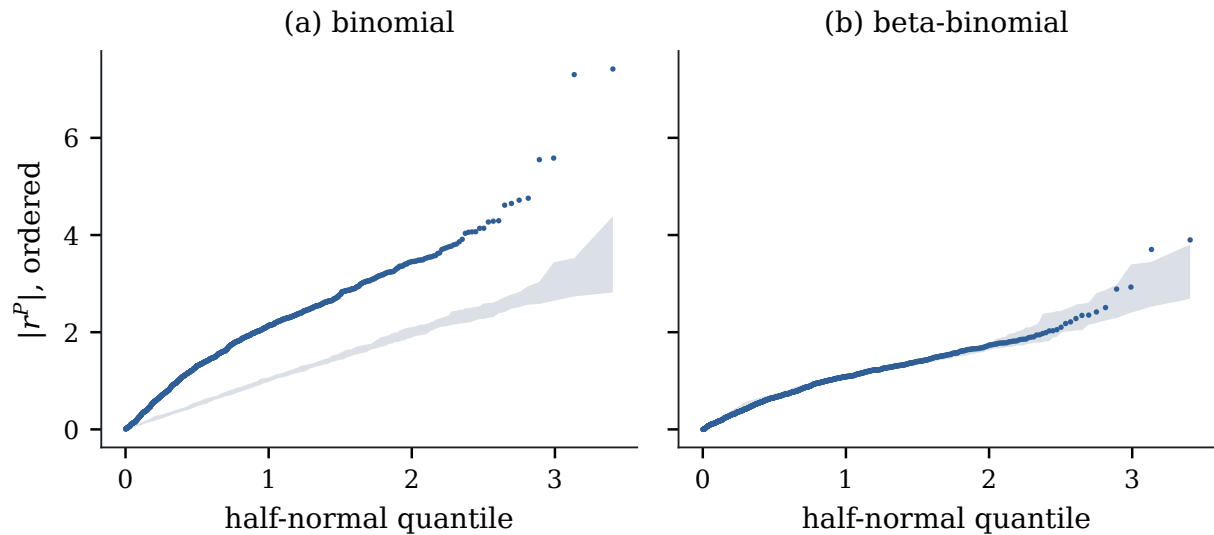


Figure 39.2.1: Half-normal plots of the absolute Pearson residuals with an envelope from 19 simulated data sets, refitted each time. (a) The binomial model: every point lies above the envelope. (b) The beta-binomial model: the upper tail is contained, the lower one is not.

and precision  $s$ , gives  $\hat{s} = 1.072$ , a within-respondent correlation of  $\rho = 1/(1+s) = 0.483$ , and the variance  $7\pi_i(1-\pi_i)(7+s)/(1+s)$ , about 3.90 times the binomial one, against the  $\hat{\phi} = 3.828$  estimated directly. Panel (b) of Figure 39.2.1 repeats the half-normal plot under that model: only 0.2 per cent of the ordered residuals now lie above the envelope, although 22.8 per cent still fall below it, against a pointwise level of 5 per cent on that side — a residual misfit at the bottom of the scale that the beta-binomial does not remove.

The consequences for inference are what Chapter 38 predicts. The age coefficient barely moves, from 0.0424 to 0.0405; its standard error moves from 0.0018 to 0.0033, and the quasi-binomial standard error — the binomial one multiplied by  $\sqrt{\hat{\phi}}$  — is 0.0034, which agrees with the beta-binomial one to two figures. The education coefficient is the casualty: 0.0233 with standard error 0.0178 under the binomial, already unconvincing at  $t = 1.3$ , becomes  $-0.0013$  with standard error 0.0343 — an effect of zero with three times the uncertainty.

```
import numpy as np
import statsmodels.api as sm
from scipy import optimize, special, stats
anes = sm.datasets.anes96.load_pandas().data
m = 7.0                                     # days in a week
y = anes["TVnews"].to_numpy(float)         # successes out of seven
X = np.column_stack([np.ones(len(y)), anes["age"], anes["educ"], anes["income"]])
n, p = X.shape

fit = sm.GLM(np.column_stack([y, m - y]), X, family=sm.families.Binomial()).fit()
mu = fit.fittedvalues * m                  # fitted expected days
eta = fit.predict(which="linear")
pearson = (y - mu) / np.sqrt(mu * (1 - mu / m))  #  $r^P$ , with  $\phi = 1$ 

print(f"deviance {fit.deviance:8.1f} on {n - p} degrees of freedom")
print(f"Pearson {np.sum(pearson**2):8.1f}; dispersion estimate "
      f"{np.sum(pearson**2) / (n - p):.3f}")
```

[Run this code online](#)

```
B = 19                                     # simulated data sets

def halfnormal_quantiles(k):
    """The horizontal axis: expected half-normal order statistics."""
    i = np.arange(1, k + 1)
    return stats.norm.ppf((i + k - 0.125) / (2 * k + 0.5))

def envelope(draw, refit_abs_resid, seed):
    """Smallest and largest of B ordered |residual| vectors from the fitted model.
    """
    rng = np.random.default_rng(seed)
    out = np.array([np.sort(refit_abs_resid(draw(rng))) for _ in range(B)])
    return out.min(axis=0), out.max(axis=0)

def binomial_abs_resid(ysim):
    f = sm.GLM(np.column_stack([ysim, m - ysim]), X, family=sm.families.Binomial()).
    fit()
    mu_s = f.fittedvalues * m
    return np.abs((ysim - mu_s) / np.sqrt(mu_s * (1 - mu_s / m)))

q = halfnormal_quantiles(n)
lo, hi = envelope(lambda rng: rng.binomial(int(m), fit.fittedvalues).astype(float),
                  binomial_abs_resid, seed=3901)
ordered = np.sort(np.abs(pearson))
outside, above = np.mean((ordered > hi) | (ordered < lo)), np.mean(ordered > hi)
print(f"binomial model: {100 * outside:.1f}% of the ordered residuals leave the
      envelope, "
      f"{100 * above:.1f}% above it")
```

[Run this code online](#)

```
# (i) is the link right? Add the squared linear predictor and test it.
fit_link = sm.GLM(np.column_stack([y, m - y]), np.column_stack([X, eta**2]),
                  family=sm.families.Binomial()).fit()
lr_link = fit.deviance - fit_link.deviance

# (ii) is the age term right? A partial residual, on the link scale.
working = (y - mu) / (mu * (1 - mu / m))          # (y - mu) dEta/dMu
partial_age = fit.params[1] * anes["age"].to_numpy() + working

# (iii) is the variance function right? Regress log (r^P)^2 on the linear predictor
lin = sm.OLS(np.log(pearson**2 + 1e-12), sm.add_constant(eta)).fit()

print(f"link test: deviance drop {lr_link:.2f} on 1 d.f., p = {stats.chi2.sf(lr_link
, 1):.3f}")
print(f"slope of log (r^P)^2 on the linear predictor: {lin.params[1]:.3f} "
      f"(t = {lin.tvalues[1]:.2f})")
```

[Run this code online](#)

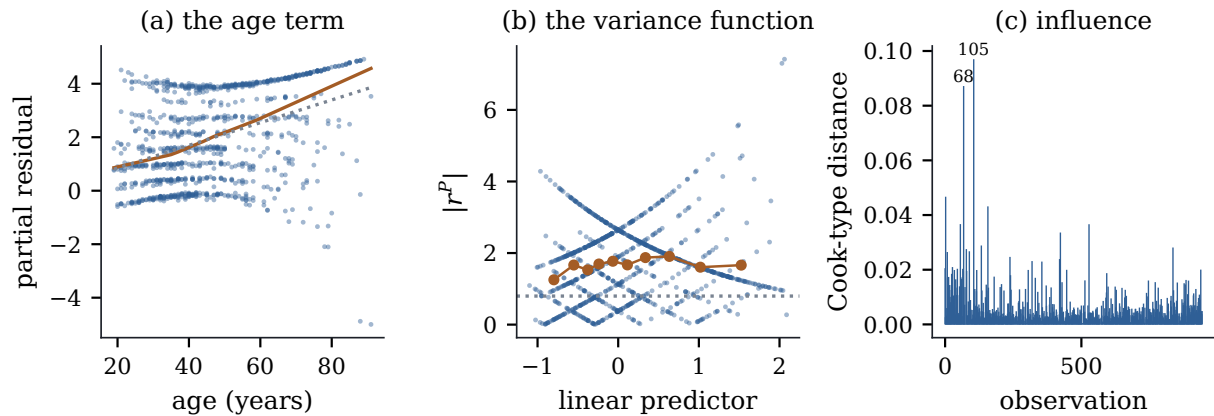


Figure 39.2.2: Three targeted checks of the binomial fit. (a) Partial residual for age, with a smooth (solid) and the fitted linear term (dotted). (b) Absolute Pearson residuals against the linear predictor, with their means in ten equal bins and the reference height  $\sqrt{2/\pi}$ . (c) Cook-type distances. Only (b) shows a problem.

```
scale = np.array([1.0, X[:, 1].std(), X[:, 2].std(), X[:, 3].std()]) # keep BFGS
                             conditioned

def betabinom_fit(ysim, start):
    """Maximize the beta-binomial likelihood: logit mean X beta, precision s."""
    def negll(theta):
        pi = special.expit((X / scale) @ theta[:p])
        s = np.exp(theta[p])
        v = -np.sum(special.betaln(ysim + s * pi, m - ysim + s * (1 - pi))
                    - special.betaln(s * pi, s * (1 - pi)))
        return v if np.isfinite(v) else 1e12

    theta = np.append(start[:p] * scale, start[p])
    for _ in range(3): # restart until the
        # gradient is flat
        opt = optimize.minimize(negll, theta, method="BFGS")
        theta = opt.x
    return np.append(theta[:p] / scale, theta[p]), opt

theta_bb, opt = betabinom_fit(y, np.append(fit.params, 0.1))
beta_bb, s_bb = theta_bb[:p], np.exp(theta_bb[p])
rho = 1.0 / (1.0 + s_bb) # correlation within the
                           week
print(f"beta-binomial: s = {s_bb:.3f}, within-responent correlation {rho:.3f}")
for j, name in enumerate(["intercept", "age", "educ", "income"]):
    print(f" {name:10s} binomial {fit.params[j]:8.4f}   beta-binomial {beta_bb[j]
    :8.4f}")
```

[Run this code online](#)

### 39.2.3 Which repair

[Example \(Days of television news\)](#) had three candidate repairs and took the third; the choice among them is the subject of Proposition 38.4.1. A **quasi-likelihood** fit keeps the mean model and the shape of the variance function and lets  $\phi$  float (Definition 38.1.1): it changes no coefficient, because the estimating equations are the same, and multiplies every standard error by  $\sqrt{\hat{\phi}}$ . A **mixture** model, such as the beta-binomial here or the negative binomial of Definition 37.4.1, names a mechanism and is a genuine likelihood, so it has an AIC and a likelihood ratio test, and it generally does move the coefficients. A **random effect** in the linear predictor is the third route, developed in [Chapter 40](#); it changes the meaning of the coefficients as well as their values (Proposition 40.1.1).

*Idea.* Read diagnostics for a generalized linear model as three separate questions. Is the mean right? Is the link right? Is the variance function right? A plot that fails is useful only when it belongs to one of the three.

### 39.2.4 Exercises

#### 39.2.4.1 A. Check your understanding

##### Exercise A1.

[Proposition 39.2.1\(b\)](#) gives the pointwise level  $2/(B+1)$ . What is  $B$  for a level of 5 per cent? If instead the envelope is taken to be the second smallest and second largest of  $B$  simulated values, what is the pointwise level?

*Solution.*  $2/(B+1) = 0.05$  gives  $B = 39$ . With the second order statistics as limits the observed value must be among the two smallest or two largest of  $B+1$ , which has probability  $4/(B+1)$ , so  $B = 79$  is needed: a less jagged envelope at the cost of more simulation.

##### Exercise A2.

The link test of [Proposition 39.2.1\(d\)](#) came out non-significant in [Example \(Days of television news\)](#). Is that evidence that the logit link is correct? What would a significant result have shown?

#### 39.2.4.2 B. Practice

##### Exercise B1.

Show that in a normal linear model with  $\phi = \sigma^2$  and the identity link, [Proposition 39.2.1\(a\)](#) reduces to the exact statement  $\text{Cov}(\hat{\epsilon}) = \sigma^2(\mathbf{I} - \mathbf{M})$  of Eq. 20.1.2, and that the standardization by  $\sqrt{1 - h_{ii}}$  is then exact rather than approximate.

*Solution.* With the identity link and constant variance  $h'(\eta) = 1$ ,  $V(\mu) = 1$ ,  $w_i = 1$ , so  $\mathbf{W} = \mathbf{D} = \mathbf{I}$  and  $\mathbf{H} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top = \mathbf{M}$ . The expansion in the proof becomes an identity,  $\hat{\boldsymbol{\mu}} - \boldsymbol{\mu} = \mathbf{M}(\mathbf{y} - \boldsymbol{\mu})$ , whence  $\text{Cov}(\mathbf{y} - \hat{\boldsymbol{\mu}}) = \sigma^2(\mathbf{I} - \mathbf{M})$  and  $r_i^P = (y_i - \hat{\mu}_i)/\sigma$  has

variance exactly  $1 - h_{ii}$ .

### Exercise B2.

Let  $Y \mid \Pi \sim \text{Binomial}(m, \Pi)$  with  $\Pi \sim \text{Beta}(s\pi, s(1 - \pi))$ . Show that  $E(Y) = m\pi$  and  $\text{Var}(Y) = m\pi(1 - \pi)(m + s)/(1 + s)$ , and that the ratio to the binomial variance is  $1 + (m - 1)\rho$  with  $\rho = 1/(1 + s)$ . Evaluate at  $m = 7$  and the fitted  $\hat{s} = 1.072$ .

*Solution.*  $E\Pi = \pi$  and  $\text{Var}\Pi = \pi(1 - \pi)/(1 + s)$  for that beta distribution. By the tower property  $EY = mE\Pi = m\pi$ , and

$$\text{Var}Y = E\{m\Pi(1 - \Pi)\} + \text{Var}(m\Pi) = m\{\pi - E\Pi^2\} + m^2 \text{Var}\Pi.$$

With  $E\Pi^2 = \text{Var}\Pi + \pi^2$  this is  $m\pi(1 - \pi) - m\text{Var}\Pi + m^2 \text{Var}\Pi = m\pi(1 - \pi)\{1 + (m - 1)/(1 + s)\}$ , which is the stated formula since  $1 + (m - 1)/(1 + s) = (m + s)/(1 + s)$ . The ratio to  $m\pi(1 - \pi)$  is  $1 + (m - 1)\rho$ . At  $m = 7$  and  $s = 1.072$ ,  $\rho = 0.483$  and the ratio is  $1 + 6 \times 0.483 = 3.90$ , close to the  $\hat{\phi} = 3.828$  that [Example \(Days of television news\)](#) estimates directly.

### 39.2.4.3 C. Going deeper

#### Exercise C1.

[Proposition 39.2.1\(b\)](#) says the simultaneous level of an envelope exceeds the pointwise one. Construct a simultaneous band from the same  $B$  simulations, by computing for each simulated data set the maximum over  $k$  of the standardized deviation  $\{R_k^{(b)} - \bar{R}_k\}/\text{sd}_k$  and using its  $(1 - \alpha)$  quantile. Show that the resulting band has simultaneous level at most  $\alpha$  under the same exchangeability assumption, and say what it costs.